Geometry : Vectors — problem 12.1.41

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Given any box $ABCDEFGH$. Prove the following theorems :

The sum of the squares of the space diagonals is four times the sum of the squares of the of the three edges.

The square of a space diagonal starting in some vertex is the sum of the squares of the face diagonals which start at the same point minus the sum of the squares of the three edges.

The sum of the lengths of a space diagonal starting at some point and the edges is greater then the sum of the face diagonals starting at the same point.

$|\vec{a}+\vec{b}+\vec{c}|+|\vec{a}|+|\vec{b}|+|\vec{c}|>|\vec{a}+\vec{b}|+|\vec{b}+\vec{c}|+|\vec{c}+\vec{a}|$ (ATMO 1972).